UTM UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SEMESTER II 2025/2026

PROBABILITY AND STATISTICAL DATA ANALYSIS

SCST1223

ASSIGNMENT 1 (50 MARKS)

INSTRUCTIONS:

1. This is a pair assignment (2 members in a group). Please clearly write name of members and matric numbers on the front page of submission.
2. Assignment 1 contributes to **5%** of overall course marks.
3. Only **HANDWRITTEN** submission is **ACCEPTED** :
    a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc.,) will be **REJECTED**.
    b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
    c. For answers that need to draw graphs, using graph paper(s) is optional. You can use plain paper.
    d. Round your answers to **TWO** decimal places.
    e. Please scan/snapshot your work and save it as a **PDF** file.
4. Submission via eLearning, only **ONE** group member needs to submit on behalf of your pair.

Question 1 **(15 Marks)**

The Faculty of Computing (FC), Universiti Teknologi Malaysia (UTM), has 450 students registered in the year 2025. To better understand students' study habits, the faculty conducted a small survey by selecting 5 students from the registered students (Year 1-Year 4). The data collected from these students are presented in **Table 1**.

Table 1: Survey of study habits

Student ID	Year of Study	Preferred Study Location	Study Hours /Day	Number of Courses	GPA	Satisfaction Level (1–5)
S01	Year 1	Library	3.5	5	3.20	4
S02	Year 2	Home	2.0	4	2.85	3
S03	Year 3	Cafe	4.0	5	3.60	5
S04	Year 1	Hostel	1.5	4	2.70	2
S05	Year 4	Library	3.0	6	3.75	4

(a) Based on the given study,
   i. Identify the population of this study.
   ii. Identify the sample used in this study. **(2 marks)**

(b) From the dataset above, classify each variable as qualitative (categorical) or quantitative (numerical). **(4 marks)**

(c) Among the quantitative variables in 1(b), determine which variables are discrete and which are continuous. Provide justification. **(4 marks)**

(d) Determine the level of measurement (nominal, ordinal, interval, or ratio) for each variable. Complete the **Table 2**. **(5 marks)**

Table 2: Level of measurement each variable

Variable	Level of Measurement	Justification
Student ID		
Year of Study		
Preferred Study Location		
Hours Studied per Day		
Number of Courses		
GPA		
Satisfaction Level		

Question 2 **(10 Marks)**

A fitness centre recorded the time spent exercising (in minutes) by 40 members for one evening session. The data were grouped as shown in **Table 3**.

Table 3: Fitness centre exercise records

Exercise Time (minutes)	Frequency
10 – 19	5
20 – 29	8
30 – 39	12
40 – 49	9
50 – 59	6

(a) Complete the following **Table 4**. **(3 marks)**

Table 4: Fitness centre exercise frequency records

Exercise Time (minutes)	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative Frequency
10 – 19	5			
20 – 29	8			
30 – 39	12			
40 – 49	9			
50 – 59	6			

(b) Construct a histogram to represent the frequency distribution for your answer in (a). **(3 marks)**

(c) Construct a relative ogive graph based on your answer in (a) **(3 marks)**

(Note: Label all axes/sections clearly for both (b) and (c).)

(d) Based on the graph in (b) and (c), describe the overall pattern of exercise time among the members. **(1 mark)**

Question 3 **(15 Marks)**

A university cafeteria recorded the amount of money spent (in RM) by 20 students during lunch. At the same time, the cafeteria also recorded the type of meal purchased by each student as shown in **Table 5**.

Table 5: Cafeteria lunch data

Student	Amount Spent (RM)	Meal Type	Student	Amount Spent (RM)	Meal Type
S1	8	Rice	**S11**	18	Western
S2	12	Noodles	**S12**	12	Noodles
S3	10	Rice	**S13**	9	Rice
S4	15	Western	**S14**	14	Western
S5	9	Rice	**S15**	10	Rice
S6	14	Noodles	**S16**	17	Western
S7	11	Rice	**S17**	11	Noodles
S8	16	Western	**S18**	8	Rice
S9	13	Noodles	**S19**	13	Noodles
S10	7	Rice	**S20**	15	Western

(a) Complete **Table 6** for the frequency distribution of meal type. **(2 marks)**

Table 6: Frequency distribution of meal type

Meal Type	Frequency	Percentage (%)
Rice		
Noodles		
Western		
Total		

(b) Construct a pie chart to represent the distribution of meal types. Clearly label each section with the meal category and percentage. **(3 marks)**

(c) Calculate the minimum, first quartile (Q1), median, third quartile (Q3), interquartile range (IQR), and maximum based on the recorded amount spent (RM). **(5 marks)**

(d) Using the 1.5 × IQR rule, determine whether any outliers exist in the dataset. **(1 mark)**

(e) Draw a modified box plot showing outliers (if any). **(4 marks)**

Question 4 **(10 Marks)**

A retail company wants to analyse its annual sales revenue (RM) over the past 12 years to understand the overall sales pattern and identify whether extreme values exist in the data as shown in Table 6.

Table 6: Annual sales revenue

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Sales	120	125	123	128	130	132	132	129	131	134	180	210

(a) Calculate mean, median and mode of the sales revenue. **(3 marks)**

(b) Calculate the variance and standard deviation. **(3 marks)**

(c) Calculate the kurtosis of the dataset. Based on your result, classify the distribution and explain what this implies about the company's sales pattern. **(2 marks)**

(d) The company recorded unusually high sales in recent years.
Which statistical measure is more suitable to represent typical sales performance? Explain your answer. **(2 marks)**

-End of Assignment-